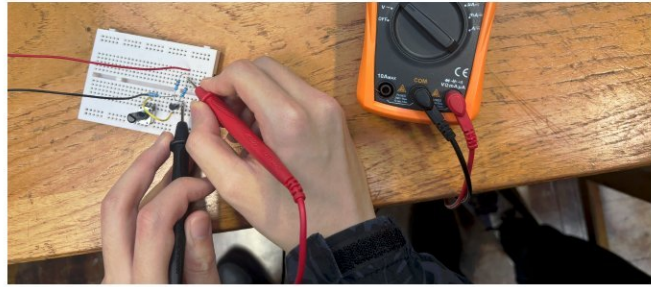
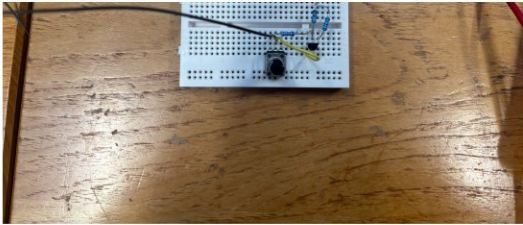


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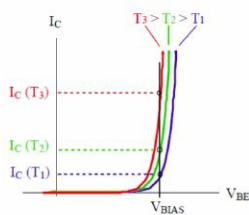
Now that the circuit is properly biased let's test how sensitive it is to temperature changes: carefully hold the transistor's plastic package between your fingers (whilst it is powered up). Observe how the collector voltage changes. Make a note of the minimum collector voltage you achieve. Can you explain why this happens?

Holding BJT between fingers causes temperature of BJT to rise, according to:

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

Where $V_T = KT/q$, it causes I_C to decrease, I_S doubles for every 8 degrees temperature increase, it causes I_C to increase and outweigh the effect of V_T .

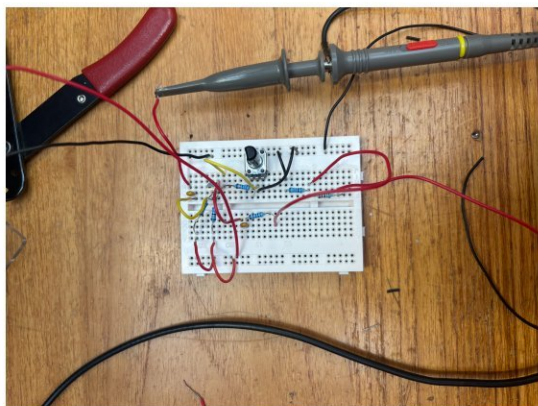
So overall, I_C increases. $V_{out} = V_{CC} - V_C = V_{CC} - I_C R_C$, thus V_{out} decreases.



The R_{B3} measured is 8.93k Ω , although it is not in range of calculation, it is quite close. Guess for the reason is because of the environment. E.g. an estimated, not accurate value of V_{BE} , temperature, and beta.

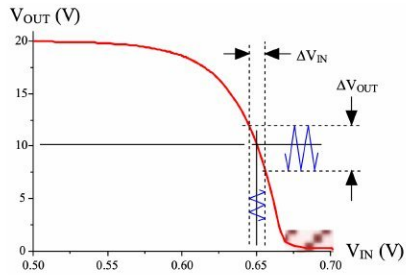
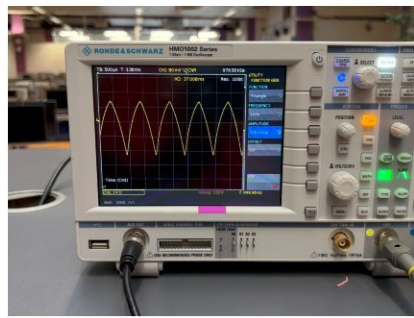
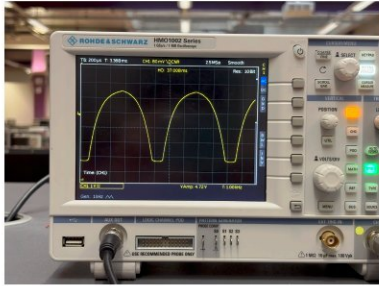
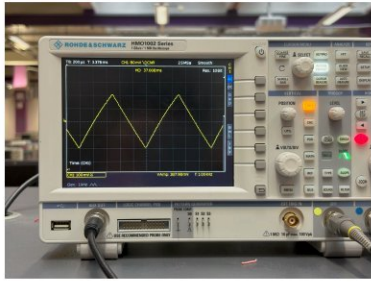
Amplifier performance

Add C_1 and C_2 to provide AC coupling. It prevents any DC voltage from the input changing the biasing and raising V_{BE} above 0.7V. Connect the signal generator to V_{in} and the oscilloscope probes to V_{in} and V_{out} .



When $V_{in} = 1\text{kHz}$ and 5mV. You should see V_{out} showing the same signal with a much larger amplitude. Increase the amplitude of V_{in} until V_{out} starts to show distortion. Record what the

distortion looks like and find the maximum output amplitude without distortion. What happens if the bias point of V_C is adjusted slightly away from 2.5V by changing R_{B3} ?



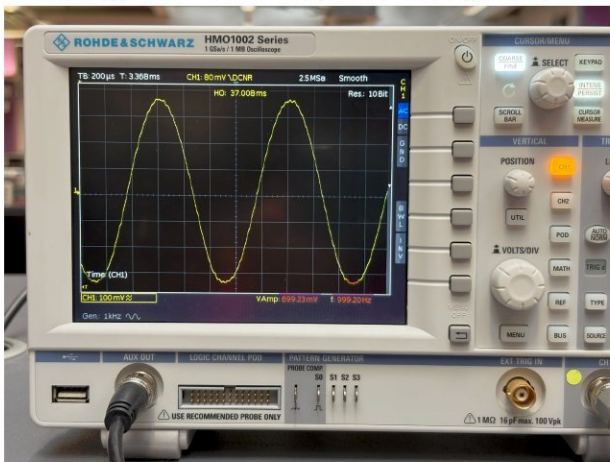
Initially it amplifies signals well. When we increase V_{in} the signals curved because the BJT will only have a linear relationship in a certain range of voltage (shown in diagram). Outside the region $V_{out} = V_{cc} - I_{cR_c}$ shows some exponential relationships, thus curved.

Changing the bias will cause the offset of signals changes, and will cause a unsymmetric distortion if distortion happens.

Maximum amplitude got is 3.56V

Switch to a sine wave with an amplitude that causes no distortion and find the gain of the amplifier by calculating V_{OUT}/V_{IN} .

- Find the gain and maximum output amplitude of your amplifier.



Maximum amplitude is 699.23mV

V_{in} is $(539mV/2)$ with a potential divider of 1kohm & 11kohm = 22.46mV

Gain = $699.23/22.46 = 31.1322$

Find the input impedance of the amplifier by deriving:

$$V_{OUT1} = V_{OUT2} \left(1 + \frac{10k}{R_{IN}}\right)$$

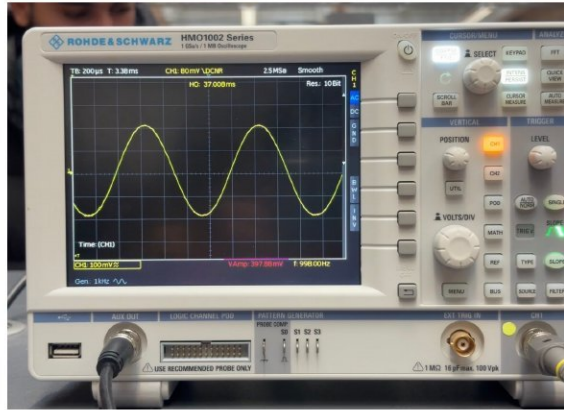
Here, R_{IN} is the input impedance of the amplifier. Can you derive this formula yourself?

It is just an equation of potential divider, where

$$\frac{V_{OUT1}}{V_{OUT2}} = \frac{AV_s}{AV_s \cdot \frac{R_{IN}}{R_{IN}+10k}} = \frac{R_{IN} + 10k}{R_{IN}} = 1 + \frac{10k}{R_{IN}}.$$

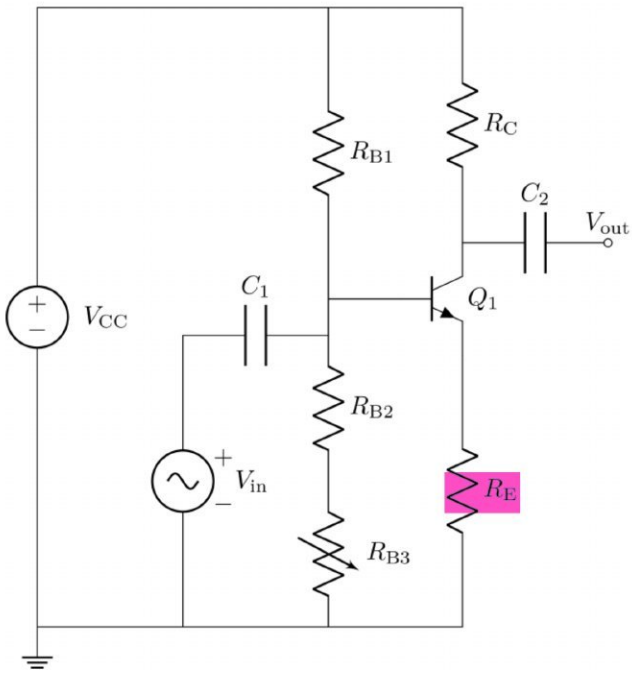
$$V_{OUT1} - V_{OUT2} = 699.23 - 397.88 = 301.35$$

$$R_{IN} = 10\,000 \cdot \frac{397.88}{301.35} \approx 10\,000 \cdot 1.320 \approx 13\,200\ \Omega$$



Emitter Degeneration

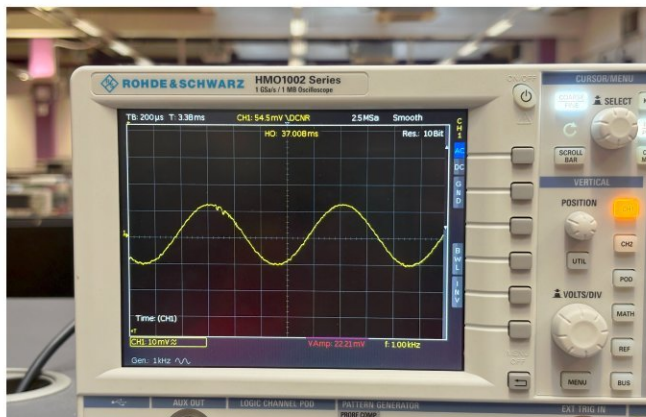
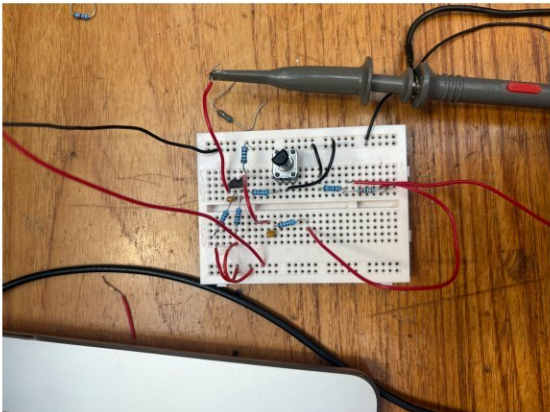
This circuit amplifies but is unstable and changes when temperature changes.
Now add emitter resistor R_E



Build the amplifier with emitter degeneration, neglect C_1 , C_2 first and set up the biasing.
Assess the temperature stability.

The reading nearly doesn't change and temperature stability is much better than before.

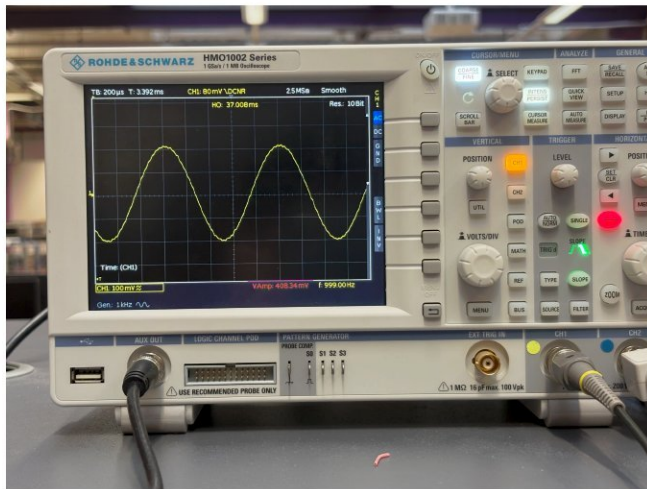
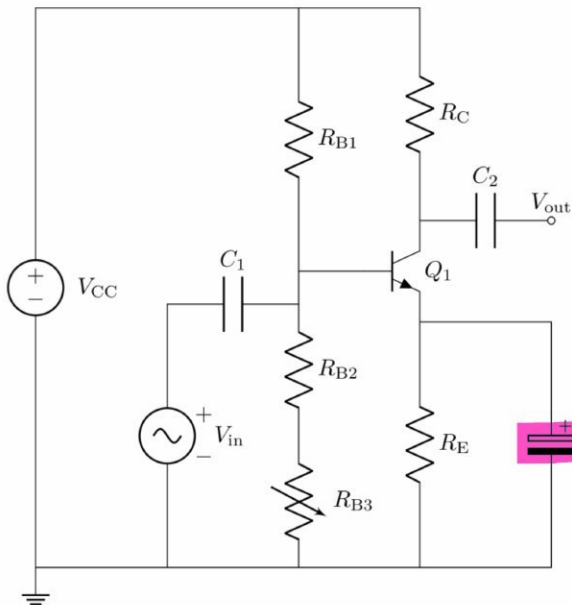
Now add C_1 and C_2 , and connect the Oscilloscope. Repeat the measurements of maximum output amplitude and gain. How do they compare to the CE amplifier without emitter degeneration?



$V_{out} = 22.21\text{mV};$
 $V_{in} = (60\text{mV}/2)/12 = 2.5\text{mV};$
 So gain = $V_{out}/V_{in} = 8.884$, which is much smaller than before.



The circuit below uses an additional electrolytic capacitor in parallel to the emitter resistor. Add it to your circuit and find the effect on gain. Biasing won't be affected because the capacitor has no impact on DC voltages or current.



$V_{out} = 408.34\text{mV};$
 $V_{in} = (60\text{mV}/2)/12 = 2.5\text{mV};$
 So gain = $V_{out}/V_{in} = 163.3$.

Gain increases a lot and is again in the range of 110-800.
 So the performance is much better.